Yihan Zhang
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RESEARCH INTERESTS

Personalized Longitudinal Monitoring · Wearable & Home Health Sensing · Trustworthy Clinical AI · Irregular & Incomplete Health Time Series · Uncertainty Quantification & Calibration

EDUCATION

MSc **Nanyang Technological University**, Singapore
Expected July 2027 | **MSc in Artificial Intelligence in Medicine** *— Lee Kong Chian School of Medicine*

BS **The University of North Carolina at Chapel Hill**, NC, USA
Dec 2025 | **Information Science; Minors in Studio Art & Geography**

SELECTED COURSEWORK

NTU (in progress): Machine Learning for Healthcare AI · Deep Learning · Human Anatomy & Physiology

MANUSCRIPTS

Yihan Zhang. *Finite-Population Reference Audits for Model Comparison.*
Submitted to AISTATS 2027, 2026. ***Sole author.***

- Developed a finite-population audit for fixed model comparisons under partial access to a second reference; proved simultaneous coverage of all legal completions of unavailable answers using hypergeometric bounds and loss-displacement supports.
- In a constructed-margin study of 128 independent draws at 30% disclosure, increased correct determinations by 27.34 percentage points (95% paired CI: 25.39–29.30) over the tested direct empirical-Bernstein adaptation; gains were stable confirmations.
- Evaluated CGMacros, VTaC, Collab-CXR, and ISRUC-Sleep reference archives; extended the framework to finite-label contrasts and characterized efficiency limits from reference ambiguity, discrepancy placement, and risk allocation.

Yihan Zhang. *From Recorded Corrections to Oracle Claims: A Measurement Audit for EHR Concept Interventions.*
Submitted to the 15th IEEE International Conference on Healthcare Informatics (ICHI 2027), 2026. ***Sole author.***

- Identified a measurement-validity problem in EHR concept interventions while stress-testing C3B: replacing learned concepts with recorded clinical references measures the model's response to those records, but does not by itself identify the effect of replacing them with study-defined true concepts.
- Formalized the observed-reference versus oracle-effect distinction, constructed observationally indistinguishable data-generating worlds with opposite oracle Brier effects, and derived finite-sample certificates using independent error calibration, exact binomial count bounds, and signed worst-case loss allocation.
- Evaluated the framework on 240 frozen synthetic claims with two prediction heads and audited frozen MIMIC-IV and eICU replays. All 53 strong-effect claims per head were certified in the synthetic evaluation with no observed false approvals; the EHR audits identified the additional measurement evidence required for oracle interpretation.

RESEARCH EXPERIENCE

C3B: Auditable Concept-Decomposable Modeling of Irregular Clinical Time Series
NTU LKCMedicine Research Project | iAIM 2026 Poster Accepted | Sole Author

- Developed C3B on MIMIC-IV, a compact irregular-time model whose risk logit is exactly decomposable into eight named clinical concept contributions; in a pre-specified five-seed 12h→6h infection/SOFA benchmark (71,692 windows; 1.4% prevalence), C3B exceeded matched XGBoost on AUPRC (+0.0127; 95% patient-cluster CI [+0.0068, +0.0189]) and AUROC (+0.070; [+0.052, +0.088]).
- Built a source-traced temporal evaluation pipeline over 74,579 ICU stays separating event time from knowledge time; stress tests across endpoint definitions, external transport, missingness, and imputation exposed ranking instability and motivated the measurement-audit work above.

SELECTED RESEARCH / SYSTEMS PROJECTS

Motion Audit: Pose-Derived Motion Measurements | Independent Research · 2026
Measurement-Chain Evaluation | Manuscript in Preparation

- Audits world reference, projection, and keypoints under shared counting and angle rules. Paired equipment changes expose task-dependent effects and endpoint masking.
- On 12 new synthetic bodies, separate angle processing with unchanged event counts improves complete success from 72.11% to 88.19% (+16.09 points; body-paired 95% CI [13.31, 18.75]); a secondary output-specific counting recipe reaches 92.59%.

CUSP: Integrated Health Information and Care Workflows | NTU Healthcare AI Innovation Project · 2026
Project Lead & System Architect | iAIM 2026 Accepted | 17-member cross-functional team

- Leading a 17-member team spanning clinical/user research, data/ML, product/UX, and regulatory/business work.
- Defined low-burden wearable and optional CGM-enhanced monitoring with longitudinal personal baselines, AI interpretation, and bounded assistance.
- Building source-aware drafts, review, and correction workflows across personal insight and professional handoffs.
- Related ECG course research examines source-specific concept measurements and calibration across reference procedures.

Pluggy: Tangible Digital-Wellbeing Intervention for Bedtime Smartphone Use | UNC Capstone · 2025
System Architect & HCI Lead | Capstone Winner (1st Place)

- Designed tangible charging and feedback to support bedtime smartphone disengagement using behavioral economics.
- Conducted mixed-method user research and translated behavioral friction points into interaction mechanics and prototype requirements.

PROFESSIONAL EXPERIENCE

China Construction Bank, Institutional Business Department | Intern · Zhejiang, China · May 2024–Aug 2024

- Participated in consumer-oriented banking product development and in the design of a Human Resources Information Management System for a major institutional client.
- Contributed to grid architecture, coding, bug fixing, documentation, and iterative testing in a team-based development process.

Weiming Environmental Protection Co., Ltd. | IT Operations Intern · Zhejiang, China · Jun 2022–Aug 2022

- Supported web-based industrial applications, including front-end functionality and backend API integration.
- Developed Python + Selenium monitoring scripts that reduced failure-response time and maintenance ticket volume by 40%.
- Compared EU and Chinese environmental regulations to inform compliance features in platform design.
- Earlier industrial R&D within a Weiming Group subsidiary: first inventor on CN209791046U and co-inventor on CN215209304U, two Chinese utility-model patents for environmental-process equipment.

LEADERSHIP & SERVICE

Class Representative, MSc Artificial Intelligence in Medicine | NTU LKCMedicine · 2026–Present

- Selected as cohort representative, serving as the official liaison bridging the inaugural/AIMed cohort, senior school leadership, and programme directors.
- Actively shaping cohort-level academic and student experience by organizing executive informal dialogues with the Dean, facilitating curriculum feedback, and resolving programmatic and campus resource bottlenecks.

HONORS

2025 Selected Exhibitor — UNC Smart Design Showcase
2024 UNC MakerFest — 2nd Most Creative Prize (Chill by U) · Dean's List — University of North Carolina at Chapel Hill (3 semesters)
2022 Let's Charrette & Sustainable Campus Design Competition — 3rd Campus Topic Prize

TECHNICAL SKILLS

Programming & Data: Python · SQL · R · Java · MATLAB
Machine Learning: PyTorch · scikit-learn · XGBoost · pandas · NumPy
Research & Engineering: Jupyter · Git · LaTeX · DuckDB
Sensing & Prototyping: Arduino · Figma · 3D Printing · Unity · ESP32 · Laser Cutting
Languages: Mandarin (Native) · English (Fluent) · Japanese (Beginner)